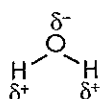


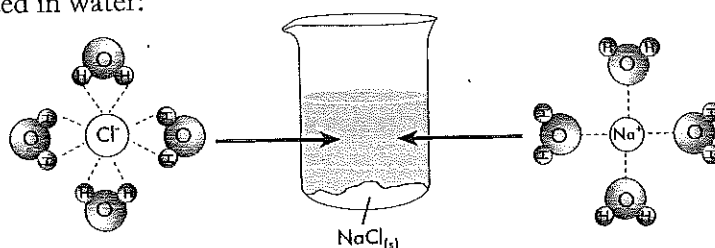
Acids and Bases in Aqueous Solution

5.1 ELECTROLYTES



The size, shape and polarity of water molecules make it possible for them to pack very closely around solute particles (ions or molecules). This makes water a very good solvent.

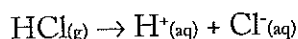
e.g. 1. $\text{NaCl}_{(s)}$ placed in water:



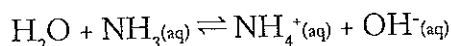
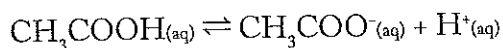
- Negative region of H_2O is attracted to positive Na^+ ion.
- H_2O molecules 'pull' Na^+ ion from the Cl^- ion and disperse the Na^+ ions throughout the solution.
- Positive region of H_2O attracted to Cl^- ions and the same process with Na^+ and Cl^- ions causes $\text{NaCl}_{(s)}$ to dissolve.

Substances that dissolve in water and form ions are called **ELECTROLYTES**.

STRONG ELECTROLYTES – Are good conductors of electricity in aqueous solution because water causes them to dissociate (ionic substances) or completely ionise (covalent molecular substances). Ionic substances are all strong electrolytes even if only slightly soluble since any amount dissolved completely breaks up into ions. Strong acids are also strong electrolytes as they completely ionise in solution.

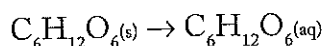


WEAK ELECTROLYTES – Are poor conductors of electricity in aqueous solution. Only a small percentage of molecules ionise when dissolved in water and so only a small concentration of ions is produced to conduct charge. Weak acids, and weak covalent molecular bases (such as NH_3), tend to be weak electrolytes.

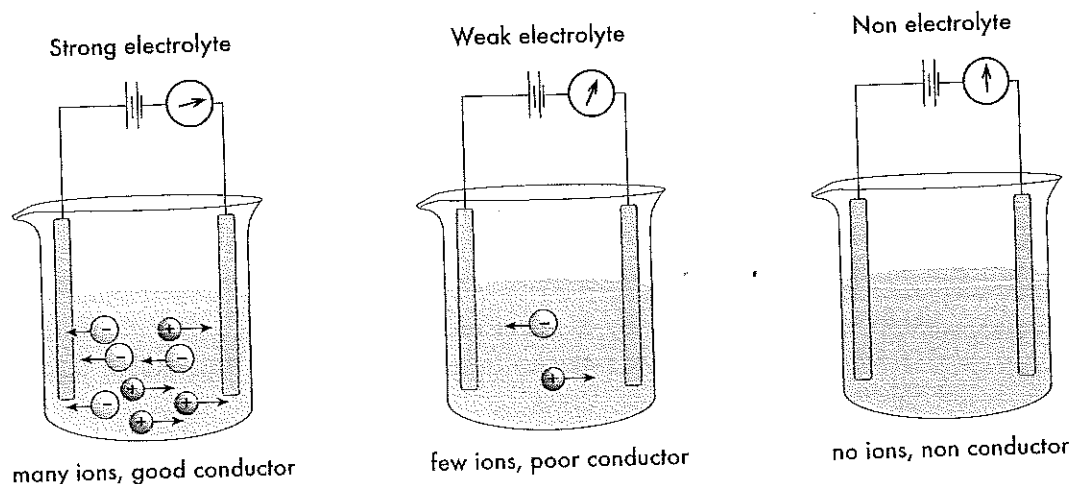


The double arrow ($\rightleftharpoons$) is used to indicate that only a small percentage of molecules ionise.

NON ELECTROLYTES – Are non-conductors of electricity because no ions are produced when they dissolve in water.

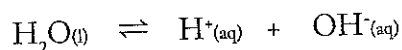


Glucose molecules dissolve but no ions are formed.



Self-Ionisation of Water

Water, no matter how pure, is always a very weak conductor of electricity. This is due to the self ionisation of water.



A 1.00 litre sample of pure water at 25°C contains 1.00×10^{-7} mol of H^+ ions and 1.00×10^{-7} mol of OH^- ions.

i.e. at 25°C, $[\text{H}^+] = [\text{OH}^-] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$ and because $[\text{H}^+] = [\text{OH}^-]$ the water is said to be neutral.

The equilibrium constant for this self ionisation is given the symbol K_w .

$$K_w = [\text{H}^+] [\text{OH}^-] = 1.00 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

The only factor that effects K_w is the temperature. As temperature rises, so does K_w but if it is pure water, no matter what the value of K_w , $[\text{OH}^-] = [\text{H}^+]$.

ACIDIC SOLUTIONS: $[\text{H}^+] > [\text{OH}^-]$

or $[\text{H}^+] > 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] < 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

BASIC SOLUTIONS: $[\text{H}^+] < [\text{OH}^-]$

or $[\text{H}^+] < 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] > 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

NEUTRAL SOLUTIONS: $[\text{H}^+] = [\text{OH}^-]$

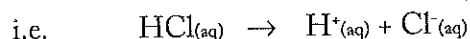
or $[\text{H}^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] = 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

Worked Examples

- 5.1 Calculate the $[\text{OH}^-]$ of a $2.00 \times 10^{-2} \text{ mol L}^{-1}$ solution of HCl

NB: HCl is a strong acid and will completely ionise



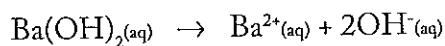
$$\text{therefore} \quad [\text{H}^+] = [\text{HCl}] = 2.00 \times 10^{-2} \text{ mol L}^{-1}$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.00 \times 10^{-14}}{2.00 \times 10^{-2}}$$

$$[\text{OH}^-] = 5.00 \times 10^{-13} \text{ mol L}^{-1}$$

- 5.2 Calculate the $[\text{H}^+]$ of a 0.350 mol L^{-1} solution of $\text{Ba}(\text{OH})_2$.



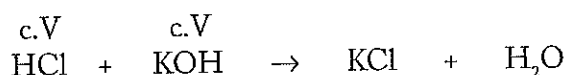
$$[\text{OH}^-] = 2 \times [\text{Ba}(\text{OH})_2] = 0.700 \text{ mol L}^{-1}$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{0.700}$$

$$[\text{H}^+] = 1.43 \times 10^{-14} \text{ mol L}^{-1}$$

- 5.3 Calculate the final pH of a solution made by mixing 25.0 mL of $0.0750 \text{ mol L}^{-1} \text{ HCl}_{(\text{aq})}$ with 32.5 mL of $0.0675 \text{ mol L}^{-1} \text{ KOH}_{(\text{aq})}$.



$$\begin{aligned} n(\text{HCl}) &= c.V = 0.0750 \times 0.0250 = 1.875 \times 10^{-3} \text{ mol} \\ n(\text{KOH}) &= c.V = 0.0675 \times 0.0325 = 2.194 \times 10^{-3} \text{ mol} \end{aligned}$$

Enough information is known to calculate the moles ($n = c.V$) of HCl and KOH. This is a limiting reagent problem.

As the mole ratio from the balanced equation is 1:1, HCl is limiting reagent as $n(\text{HCl}) < n(\text{KOH})$.

$\therefore$ In the final solution, all HCl has been consumed and the pH is affected by $n(\text{KOH})$ not consumed.

$$\begin{aligned} n(\text{KOH}) \text{ not consumed} &= 2.194 \times 10^{-3} - 1.875 \times 10^{-3} \\ &= 3.188 \times 10^{-4} \text{ mol} \end{aligned}$$

$$\begin{aligned} c(\text{KOH}) &= n/V = 3.188 \times 10^{-4} / (0.0250 + 0.0325) \\ &= 5.54 \times 10^{-3} \text{ mol L}^{-1} \end{aligned}$$

$$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{5.54 \times 10^{-3}} = 1.80 \times 10^{-12} \text{ mol L}^{-1}$$

$$\text{pH} = -\log [\text{H}^+] = 11.7$$

5.2 THE pH SCALE

The pH of a solution is simply a measure of its acidity. It depends on the hydrogen ion concentration of the solution and is defined as follows.

$$\text{pH} = -\log_{10} [\text{H}^+]$$

pH < 7	acidic
pH = 7	neutral
pH > 7	basic

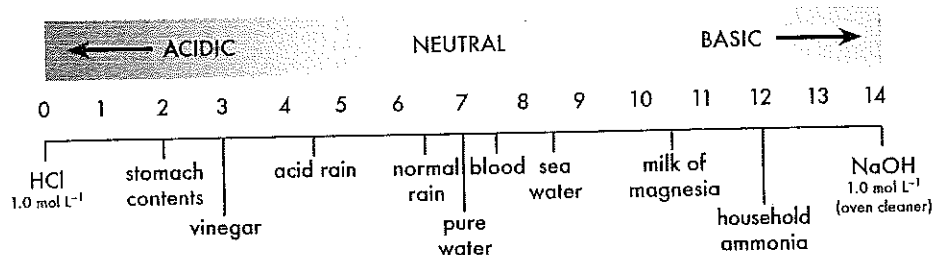


Figure 4.1 The pH acidity scale. As pH increases acidity decreases. An increase in pH by 1 means that a solution is only $\frac{1}{10}$ as acidic.

Question 5.1

Calculate the pH of 1.00 L of an antacid solution that contained 50.0 mg of $\text{Mg}(\text{OH})_2$ as its main ingredient.

Question 5.2

- (a) Determine the $[\text{H}^+]$ and the $[\text{OH}^-]$ of a sample of orange juice that had a pH of 3.60.

- (b) Determine the $[\text{H}^+]$ and $[\text{OH}^-]$ if the orange juice was then diluted by an equal volume of water (i.e. its volume was doubled using water).

Question 5.3

At 50.0°C, pure water was found to have a K_w of 5.48×10^{-14} . Determine the pH of the water at 50.0°C.

Question 5.4

A sample of sulfuric acid was found to have a pH of 2.50 at 50°C. Determine the $[\text{OH}^-]$ of this sample.

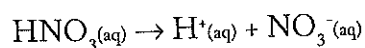
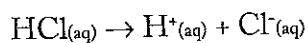
Question 5.5

Calculate the final pH of the solution formed when 47.5 mL of 0.540 mol L⁻¹ HNO₃ is mixed with 55.5 mL of 0.444 mol L⁻¹ NaOH.

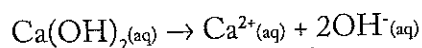
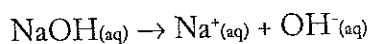
Defining Acids and Bases

Arrhenius Theory:

- acids provide hydrogen ions in aqueous solution

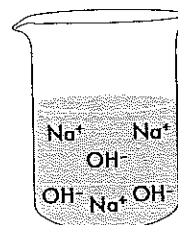
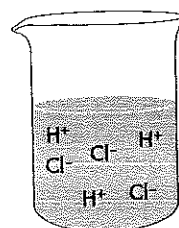
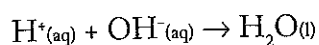


- bases provide hydroxide ions in aqueous solution



- neutralisation occurs when H^+ ion and OH^- ion combine to form $\text{H}_2\text{O}_{(\text{l})}$

Neutralisation Reaction between $\text{HCl}_{(\text{aq})}$ and $\text{NaOH}_{(\text{aq})}$ is

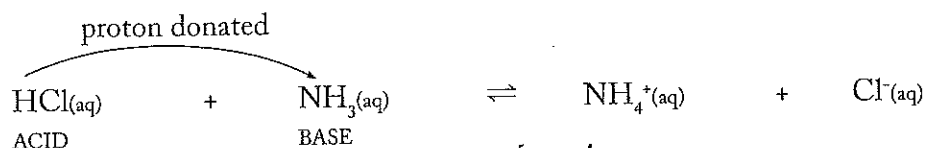


The Arrhenius model is restricted to aqueous solutions. A more general theory with wider application was proposed in 1923 by Johannes Brønsted (Denmark) and Thomas Lowry (England).

Brønsted-Lowry Theory

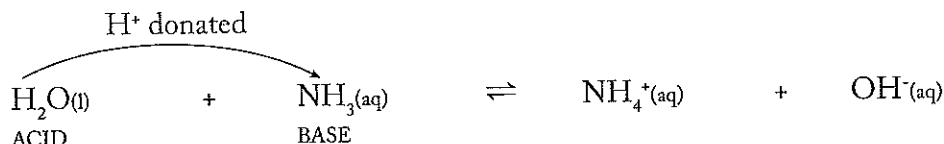
- acids act as proton (H^+) donors
- bases act as proton (H^+) acceptors

e.g. 1:

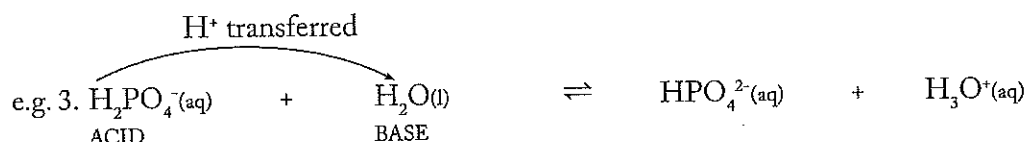


HCl is acting as an acid by donating a proton (H^+) to NH_3 . NH_3 , therefore, is acting as a base as it is accepting a proton.

e.g. 2:



H_2O is acting as an acid (proton donor)
 NH_3 is acting as a base (proton acceptor)



H_2PO_4^- is acting as an acid (donating a proton). H_2O is acting as a base (accepting a proton).

Note how H_2O can act either as an acid or a base using this theory (amphiprotic).

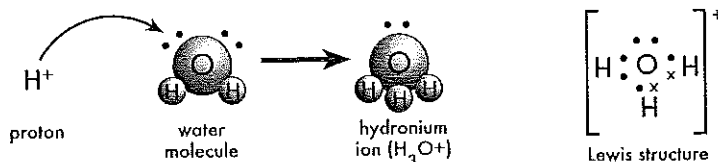
Question 5.6

Complete the following equation so as to show HCO_3^- acting as (a) an acid, (b) a base.



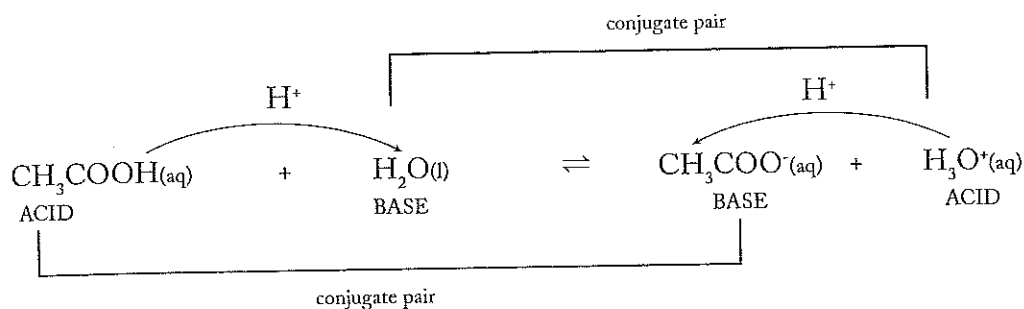
H^+ - A CLOSER LOOK!

The simplest atom, hydrogen, consists of a proton surrounded by an electron. Hence the symbol H^+ can be used to represent a proton. Protons are very small and readily attach themselves to the negative end of say, polar substances. In water they combine with H_2O to form the hydronium ion, H_3O^+ .



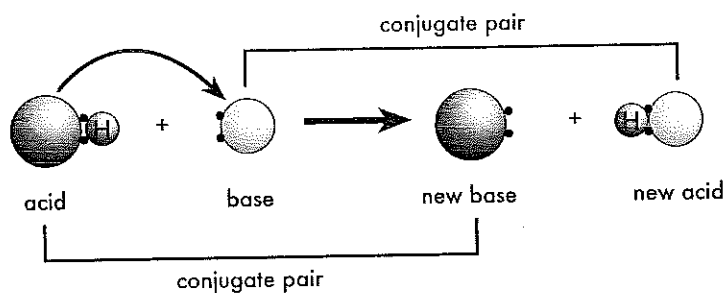
Conjugate acids and bases

When an acid loses (donates) a proton, what is left behind is its conjugate base. Similarly when a base accepts a proton, the substance formed is its conjugate acid.



i.e. $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$ and $\text{H}_3\text{O}^+/\text{H}_2\text{O}$

In general, the formation of conjugate pairs can be illustrated as follows.



Question 5.7

Complete the following tables. The first example in each case has been done for you.

(a)

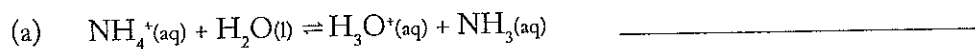
acid	conjugate base
HCl	Cl^-
H_2SO_4	
HF	
H_2S	
HCO_3^-	

(b)

base	conjugate acid
CO_3^{2-}	HCO_3^-
NH_3	
H_2PO_4^-	
F^-	
HSO_4^-	

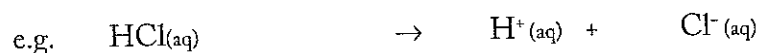
Question 5.8

Identify the conjugate acid/base pairs in each of the following.



Strength of acids and bases

Strong acids completely ionise in water. For example, hydrogen chloride molecules in solution exist essentially as hydrogen ions, H^+ and chloride ions Cl^- .



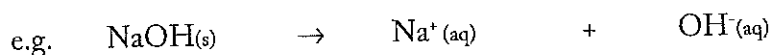
Weak acids only partly ionise in water. For example, an ethanoic acid solution contains mostly CH_3COOH molecules.



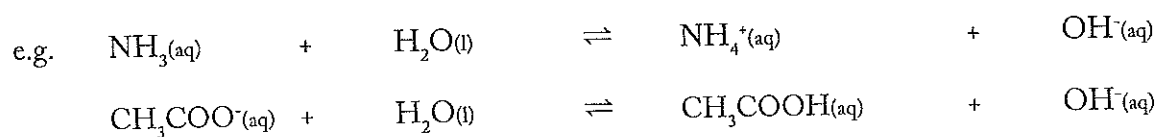
Notice the two arrows ($\rightleftharpoons$) are shown for the ionisation of weak acids.

Strong acids		Weak acids	
hydrochloric acid	HCl	oxalic acid	$H_2C_2O_4$
sulfuric acid	H_2SO_4	phosphoric acid	H_3PO_4
nitric acid	HNO_3	hydrofluoric acid	HF
hydrobromic acid	HBr	ethanoic acid	CH_3COOH
hydroiodic acid	HI	carbonic acid	H_2CO_3
perchloric acid	$HClO_4$	hypochlorous acid	HClO

Strong bases completely *dissociate* in water to produce hydroxide (OH^-) ions. Note that dissociation is simply the break up of ions that already exist within the ionic compound.



Weak bases only partially react with water in forming hydroxide (OH^-) ions. Apart from ammonia (NH_3), most bases are anions such as CO_3^{2-} and CH_3COO^- which react with water to produce OH^- ions.



Strong bases		Weak bases	
Soluble			
sodium hydroxide	NaOH	ammonia	NH_3
potassium hydroxide	KOH	carbonate ion	CO_3^{2-}
Slightly soluble		ethanoate ion	CH_3COO^-
calcium hydroxide	$Ca(OH)_2$	fluoride ion	F^-
magnesium hydroxide	$Mg(OH)_2$	phosphate ion	PO_4^{3-}

Hydrolysis of Salts

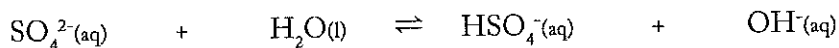
Hydrolysis is the process in which an *ion* reacts with water to produce H_3O^+ ions or OH^- ions. Ions that produce H_3O^+ are said to be acidic while ions that produce OH^- are said to be basic.

Acidic Anions:	HSO_4^- and H_2PO_4^- (i.e. anion from a polyprotic acid)
Basic Anions:	CH_3COO^- , HPO_4^{2-} , PO_4^{3-} , SO_4^{2-} , F^- , HCO_3^- , CO_3^{2-} , ClO^- , HS^- , CN^- (anion from a weak acid)
Neutral Anions:	Cl^- , NO_3^- , Br^- , I^- (anions from strong acids)
Acidic Cations:	NH_4^+ , Al^{3+} , Fe^{3+} (cation from a weak base or from aquated metal ions)
Basic Cations:	There are no basic cations covered in the syllabus
Neutral Cations:	Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Li^+ , Ba^{2+} (Group I and II metal cations)

Worked Example

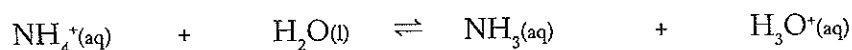
5.4 Write the equation for the hydrolysis of the following salt and classify the resulting solution as acidic or basic.

(a) Na_2SO_4 :



As OH^- ions are produced the solution is basic

(b) NH_4Cl :

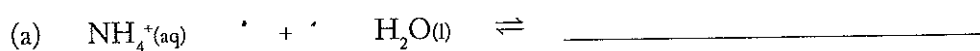


As H_3O^+ ions are produced, the solution is acidic

(NB: When writing an hydrolysis equation, the neutral anion or cation is a spectator ion and omitted)

Question 5.9

Complete the following hydrolysis reactions and indicate the type of solution they form (acidic or basic).



Solution becomes _____

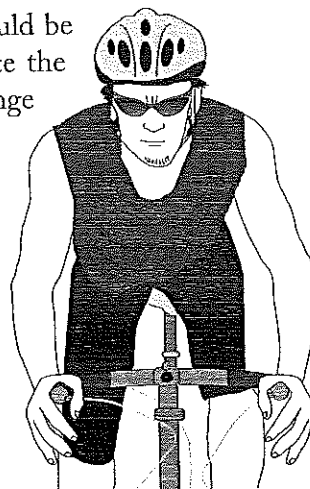
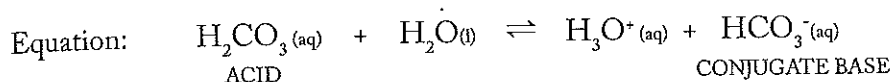


Solution becomes _____

The addition of only 0.01 mol of NaOH would neutralise 0.01 mol of H_3O^+ . The forward reaction would be favoured to partially counteract this change. As there is 0.4 mol of CH_3COOH available for the production of H_3O^+ (as per the forward reaction), the buffer would be operating within its capacity and be able to maintain a constant pH.

However if 0.5 mol of NaOH was added to the solution, the buffer would be swamped. It would not have sufficient moles of CH_3COOH to replace the H_3O^+ consumed by the NaOH. The pH of the buffer solution would change markedly.

Exercise tends to increase the acidity (reduce the pH) of blood. The presence of H_2CO_3 and its conjugate base, HCO_3^- help to maintain the bloods pH at 7.4.



During exercise more H_3O^+ is released into the blood from the muscles. Le Chatelier's Principle predicts that the reverse reaction would be favoured to partially counteract the increased $[\text{H}_3\text{O}^+]$ and effectively maintain a relatively constant blood pH.

Question 5.11

A second buffer solution in blood is the $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ combination.

(a) Write the equation for the reaction of the acidic ion, $\text{H}_2\text{PO}_4^-(\text{aq})$ with water.

(b) Identify the acid/base conjugate pairs in your equation.

(c) Use Le Chatelier's Principle to explain how this combination would respond to the addition of OH^- to the blood.

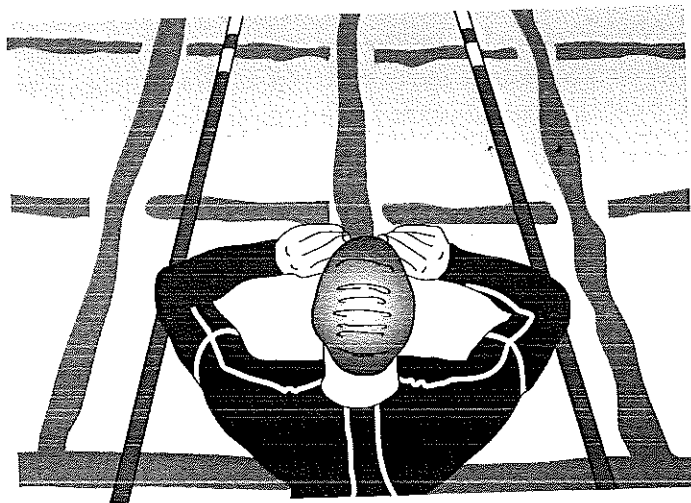
Question 5.12

Complete the following table by naming the conjugate base of the weak acids listed and then write the equation for the reaction of the weak acid with water.

WEAK ACID	CONJUGATE BASE	EQUATION
HF	F^-	$\text{HF}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$
CH_3COOH		
H_3PO_4		
NH_4^+		

Question 5.13

A 50,000 L swimming pool had a pH of 7.6.



- (a) What would be the expected new pH if 100 mL of 12.0 mol L⁻¹ HCl was added to the pool?

- (b) After adding the 100 mL of 12.0 mol L⁻¹ HCl to the pool, the owner measured the pH and found it to be 7.2. Account for the difference between the pH calculated in part (a) and the value obtained by the owner.

- (c) Suggest what chemicals could be added to the pool to form a buffer solution. Write the equilibrium equation for this buffering solution and suggest two points to consider when choosing a suitable buffer for a swimming pool.

5.3 VOLUMETRIC ANALYSIS

Volumetric analysis is a quantitative method of chemical analysis. A solution of accurately known concentration is reacted with a solution of the substance whose concentration we wish to find. By comparing the reacting volumes we can determine the concentration of the unknown.

The actual procedure of adding one solution to the other for a complete reaction is called a titration.

The two types of titrations are:

- Acid-base titrations
- Redox titrations.

The same principles apply to both and in particular the use of primary standards (see next page) to initially establish the concentration of solutions (standardisation).

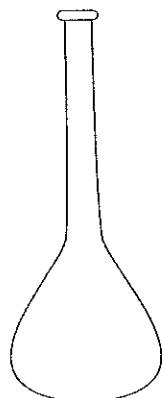
Acid-base titrations

These involve reactions between solutions of acids and bases. The point at which the reaction is complete is referred to as the equivalence point, reactants are in stoichiometric proportions. This is often not visible, as acids and bases are colourless, and so an indicator is required.

Since the products of an acid-base reaction may be acidic, basic or neutral, it is important to choose an indicator so that the end point matches the equivalence point. An appropriately chosen indicator will show a colour change (end point) when the reaction is complete (equivalence point).

Table 5.1 Choosing a suitable indicator.

Indicator	pH range	Colour change acid/base	Suitable for
methyl orange	3.1 - 4.4	red → yellow	strong acid + weak base e.g. $\text{HCl} + \text{NH}_3$
methyl red	4.4 - 6.2	red → yellow	strong acid + strong base e.g. $\text{HCl} + \text{NaOH}$
phenolphthalein	8.3 - 10.0	colourless → pink	weak acid + strong base e.g. $\text{CH}_3\text{COOH} + \text{NaOH}$



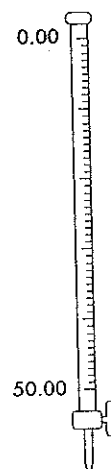
Volumetric flask
Holds accurately known volume of solution
e.g. 250.0 mL, 500.0 mL.
Rinse with distilled water.



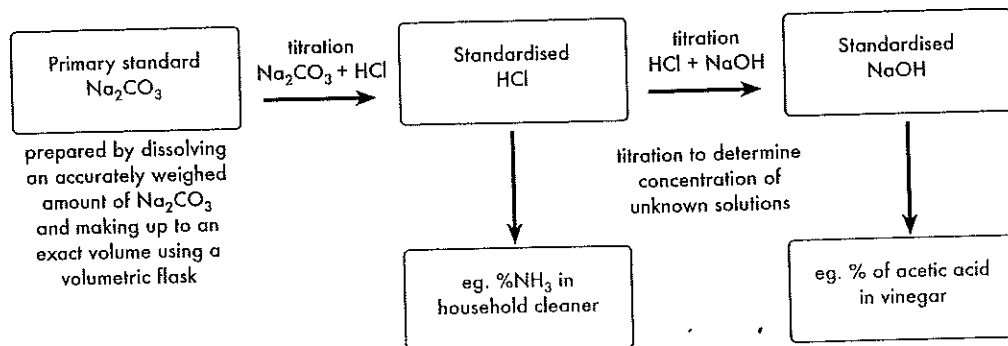
Pipette
Delivers accurately a specific volume, aliquot, of solution e.g. 20.00 mL (to ± 0.02 mL). Rinse with solution



Conical flask
Used to hold an aliquot of solution under the burette.
Rinse with distilled water.



Burette
Delivers accurately a variable volume of solution (to ± 0.02 mL).
Rinse with solution.



Primary standards

A primary standard is required to establish the concentration of unknown solutions. Some characteristics of a good primary standard are:

- high purity and known formula
- does not react with air (e.g. not hygroscopic or deliquescent)
- reasonably high molar mass
- soluble in water.

Two commonly used primary standards are anhydrous sodium carbonate (Na_2CO_3) and hydrated oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$).

Standardising solutions

The procedure involved in setting up for acid-base titrations can be illustrated as follows:

Worked Example

5.5 A student prepared a primary standard of Na_2CO_3 by dissolving 1.320 g of pure anhydrous sodium carbonate and making it up to 250.0 mL in a volumetric flask.

This solution was used to standardise some HCl acid. A 20.00 mL aliquot of the Na_2CO_3 solution required on average 21.20 mL of the acid for a complete reaction. Calculate:

- the concentration of the Na_2CO_3 solution,
- the concentration of the HCl solution.

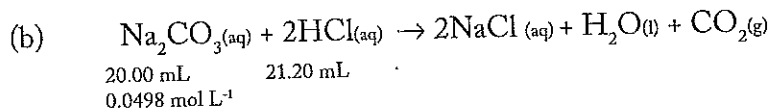
$$\begin{aligned} \text{(a)} \quad n &= \frac{m}{M} \\ M(\text{Na}_2\text{CO}_3) &= 22.99 \times 2 \\ &\quad + 12.01 \\ &\quad + 48.00 \\ &= 105.99 \text{ g mol}^{-1} \end{aligned}$$

$$\begin{aligned} n(\text{Na}_2\text{CO}_3) &= \frac{1.320}{105.99} \\ &= 0.012454 \text{ mol} \end{aligned}$$

$$c = \frac{n}{V}$$

$$\therefore c(\text{Na}_2\text{CO}_3) = \frac{0.012454}{0.2500}$$

$$= 4.98 \times 10^{-2} \text{ mol L}^{-1}$$



$$n(\text{Na}_2\text{CO}_3)_{(\text{aq})} \text{ used in the titration}$$

$$= cV = (4.98 \times 10^{-2})(0.0200) = 9.9616 \times 10^{-4} \text{ mol}$$

$$\therefore n(\text{HCl}) \text{ in } 21.20 \text{ mL} = (2)(9.9616 \times 10^{-4})$$

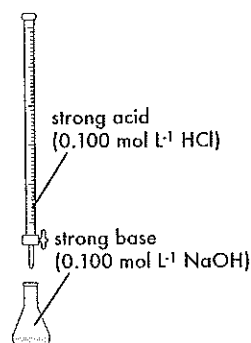
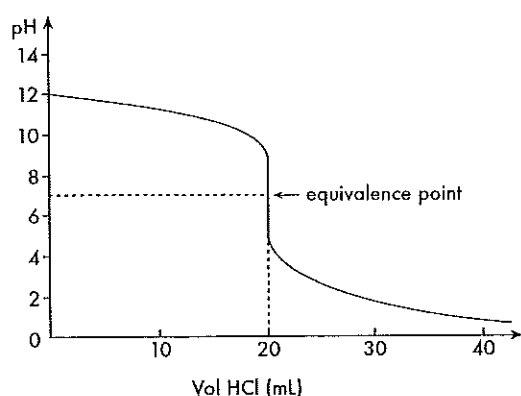
$$\text{from equation} = 1.992 \times 10^{-3} \text{ mol}$$

$$\therefore c(\text{HCl}) = \frac{n}{V} = \frac{1.992 \times 10^{-3}}{0.02120}$$

$$= 9.40 \times 10^{-2} \text{ mol L}^{-1}$$

Changes in pH during acid-base titrations

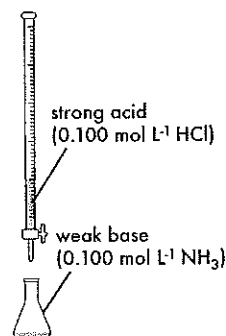
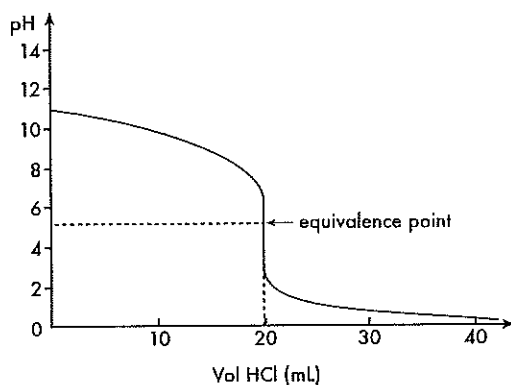
(a) Strong acid-strong base.



Equivalence point pH = 7.

Suitable indicator: Methyl orange or phenolphthalein. Methyl red often preferred.

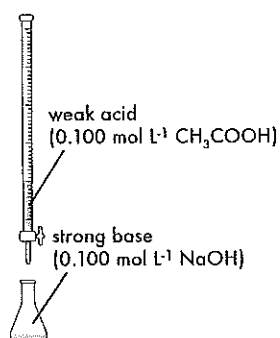
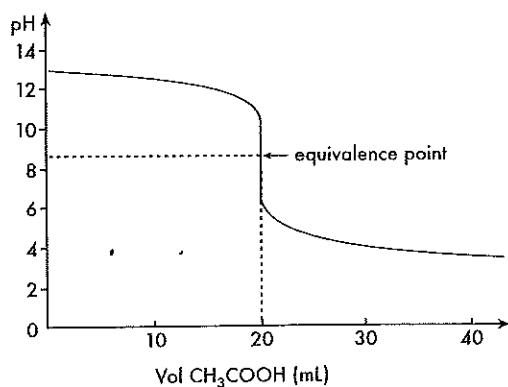
(b) Strong acid-weak base.



Equivalence point pH ≈ 5.5.

Suitable indicator: methyl orange.

(c) Weak acid-strong base.



Equivalence point pH ≈ 9.

Suitable indicator: phenolphthalein.

Question 5.14

Explain why the following substances are not suitable for use as primary standards.

(a) sodium hydroxide _____

(b) hydrochloric acid _____

Question 5.15

A student carrying out a titration between HCl acid and ammonia solution used phenolphthalein as an indicator.

(a) Is this an appropriate indicator? _____

(b) Explain clearly why/why not.

Complex problems - use flow diagrams

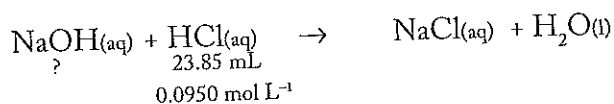
The more difficult problems are those that involve various stages or steps. It helps greatly to visualise the problem with simple sketches showing the procedure involved in the titration.

This is particularly useful when you need to work "backwards" to find percentage purity in a given sample (as in the next example).

Worked Example

5.6 A 1.252 g sample of an oven cleaner containing mostly sodium hydroxide was dissolved and made up to 250.0 mL using a volumetric flask. A 20.00 mL aliquot of this solution was titrated against $0.0950 \text{ mol L}^{-1}$ HCl and required 23.85 mL of this acid to reach the end point. Calculate the percentage (by mass) of NaOH in the oven cleaner.

Note that 1.252 g is the mass of the oven cleaner and not the mass of NaOH. In this problem we work backwards from the titration to find out how much NaOH was actually in the sample.

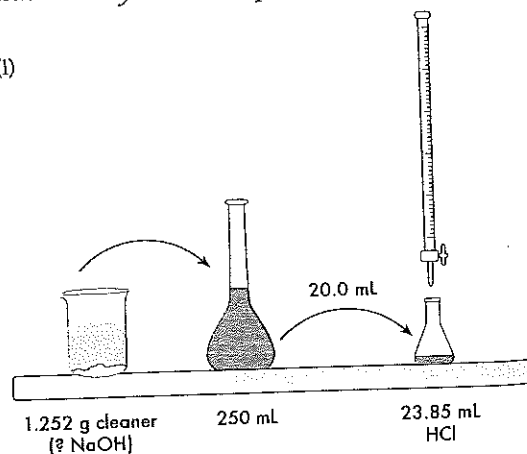


$$n(\text{HCl}) = cV = (0.0950)(23.85 \times 10^{-3})$$

$$= 2.266 \times 10^{-3} \text{ mol}$$

$$n(\text{NaOH}) \text{ in } 20 \text{ mL aliquot}$$

$$= 2.266 \times 10^{-3} \text{ mol (from equation)}$$



Now work 'backwards' to original sample.

ie. $n(\text{NaOH})$ in 250 mL

$$= (2.266 \times 10^{-3} \text{ mol}) \left(\frac{250}{20} \right)$$

$$= 2.832 \times 10^{-2} \text{ mol}$$

$\therefore m(\text{NaOH})$ in original sample

$$m = nM = (2.832 \times 10^{-2})(40.00) = 1.133 \text{ g}$$

$\therefore \% (\text{NaOH})$ by mass in original sample

$$= \frac{1.133}{1.252} \times 100 = 90.5\%$$

Question 5.16

A 15.50 g sample of commercial vinegar was diluted and made up to 250.0 mL in a volumetric flask. A 20.00 mL aliquot of this diluted vinegar required 17.75 mL of a $0.05050 \text{ mol L}^{-1}$ NaOH solution for complete neutralisation.

(a) What indicator would have been suitable for this titration? _____

(b) Determine the % by mass of ethanoic acid in the commercial vinegar sample.

(c) Determine also the concentration of ethanoic acid in commercial vinegar (in mol L^{-1} and g L^{-1}). Assume that the density of vinegar is 1.00 g mL^{-1} .

* Don't forget to draw a simple flow chart diagram to help you visualise the problem.

Question 5.17

(a) An oxalic acid solution was prepared by dissolving 0.554 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in distilled water and making it up to 500.0 mL in a volumetric flask. Determine its concentration.

- (b) This solution was placed in a burette and used to standardise a NaOH solution. A 20.00 mL aliquot of NaOH required 19.65 mL of the oxalic acid solution to reach the end point.

- (i) Name an appropriate indicator for this titration. _____
- (ii) What is the colour change at end point? _____
- (iii) Calculate the concentration of the NaOH solution.



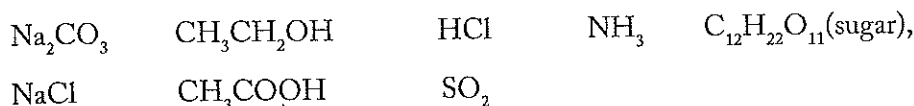
- (c) The NaOH solution was used to determine the concentration of a diluted vinegar solution. A 20.00 mL aliquot of NaOH solution required 32.50 mL of diluted vinegar solution to complete the reaction.

- (i) Name an appropriate indicator for this titration. _____
- (ii) What is the colour change at end point? _____
- (iii) Calculate the concentration of ethanoic acid (CH_3COOH) in the diluted vinegar solution.

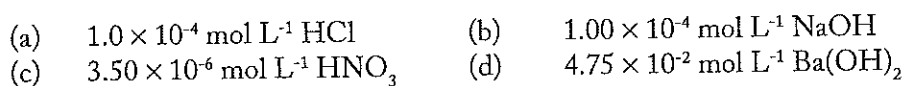
Review Questions

Chapter 5: Acids and Bases

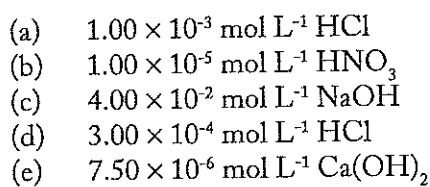
- Describe a simple experimental procedure that could be used to determine if a substance is a strong, weak or non-electrolyte. Include the comparative results for each.
- Classify the following substances as strong, weak or non-electrolytes (when they are in solution):



- Calculate the $[\text{H}^+]$ and $[\text{OH}^-]$ of the following solutions:

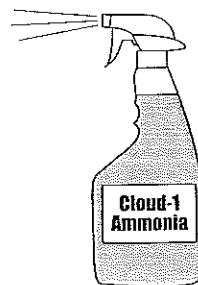


- Calculate the pH of the following solutions:



- Copy and complete the following table by calculating the $[\text{H}^+]$ and $[\text{OH}^-]$ of the household materials listed (at 25°C).

Material	pH	$[\text{H}^+]$	$[\text{OH}^-]$
vinegar	3.00		
toothpaste	6.80		
oven cleaner	13.5		
window cleaner	9.75		

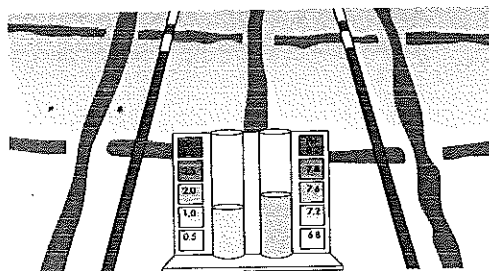


- Calculate the $[\text{H}^+]$, $[\text{OH}^-]$ and pH of the solution that is formed when 120.0 mL of a HCl solution with a pH of 4.5 is mixed with 75.0 mL of a sodium hydroxide solution with a pH of 10.2.
- A student was given 3 test tubes. One tube contained hydrochloric acid, one contained nitric acid while the third contained ethanoic acid. The three acid solutions were of equal concentration. When several drops of universal indicator were added to each test tube, the first two went red while the third turned yellow. Explain why the indicator turned a different colour in CH_3COOH .
- A sample of water was found to have a hydrogen ion concentration of $1.00 \times 10^{-7} \text{ mol L}^{-1}$ and was neutral. Explain how it is possible for the same sample of water to have a hydrogen ion concentration of $5.39 \times 10^{-8} \text{ mol L}^{-1}$ and still be considered neutral.
 - What change would have occurred to have lowered the hydrogen ion concentration from $1.00 \times 10^{-7} \text{ mol L}^{-1}$ to $5.39 \times 10^{-8} \text{ mol L}^{-1}$?

(c) If the water had a $[H^+]$ of $5.38 \times 10^{-8} \text{ mol L}^{-1}$ and was considered neutral, what would be the value of K_w .

(d) Under what conditions would K_w be greater than 1.00×10^{-14} ?

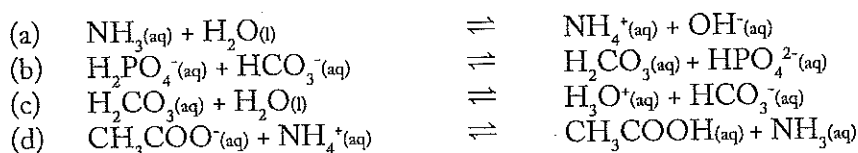
9. After having their pool water tested at the Local Pool Shop, an owner was told the pH of the water was 7.8 which was too high. To correct the problem, the owner was instructed to add 900 ml of Local Pool Shop Liquid Acid to the pool. This liquid acid was 320 gL^{-1} HCl.



(a) If the pool had a volume of 45,000 L, calculate the new pH when this quantity of Local Pool Shop Liquid Acid was added.

(b) The pH of the pool was reduced to 7.2 NOT the calculated value. Suggest a reason why the pH did not reduce to the calculated value.

10. Identify the conjugate acid/base pairs in each of the following equations:



11. Complete the following table about salt formation.

Reactants	Salt Formed	Nature of the salt (Acid, Base or Neutral)
HCl + NH ₃		
H ₂ SO ₄ + KOH		
CH ₃ COOH + NaOH		
	Mg(NO ₃) ₂	
	Na ₃ PO ₄	

12. Classify the following salts as acidic, basic or neutral and use the concept of hydrolysis to write equations to show why the salt has been classified as acidic or basic.

- ammonium nitrate
- sodium chloride
- sodium sulfide
- potassium ethanoate

13. When the weak base, Na₂CO₃ is reacted with the strong acid, HCl, the salt produced is NaCl. Na⁺ ions and Cl⁻ ions are both neutral but the resulting solution when Na₂CO₃ neutralises HCl is slightly acidic. Use equations to explain why.

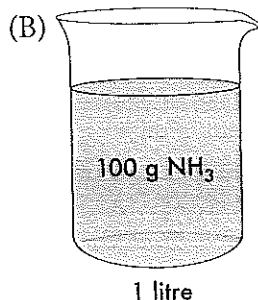
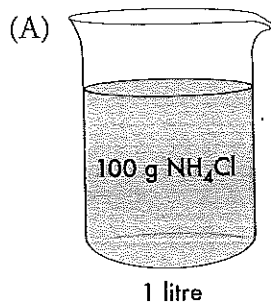
14. Give a simple test that could be used to differentiate between the following substances (give a clear statement of what observations would be expected).

- (a) 0.500 mol L⁻¹ solutions of barium hydroxide and ammonia
- (b) sodium ethanoate and sodium chloride powders
- (c) 2.00 mol L⁻¹ solutions of HCl and H₂SO₄

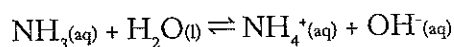
15. Rank the following 0.1 mol L⁻¹ solutions in order of increasing pH:

NaCH₃COO; NaCl, NaOH, NH₄Cl, HCl

16.



- (a) What would you expect the approximate pH of the solutions in (A) and (B) to be (>7, <7, =7). Use equations to justify your answer.
- (b) When (A) and (B) are mixed together, a buffer solution is formed that has a pH of approximately 9.



Apart from H₂O and Cl⁻, rank the chemical species in order of concentration.

- (c) When a small amount of H⁺(aq) ions are added to this buffer solution, explain why they are more likely to react with NH₃(aq) than OH⁻(aq).
- (d) Use Le Chatelier's Principle to predict what the effect would be of adding the following (separately) to this buffer solution.
 - (i) 2 g of HCl
 - (ii) 2 g of NaOH
- (e) How could the buffer capacity of this mixture be exceeded?

Titration

17. The accurate determination of the concentration of a hydrochloric acid solution, using a sodium carbonate solution of known concentration is a common analytical task carried out by high school chemistry students. Explain the meaning of the following terms used in describing titrations:

- | | |
|----------------------|-----------------------|
| (a) primary standard | (d) equivalence point |
| (b) standardisation | (e) indicator |
| (c) end point | (f) rough titration. |

18. Describe four characteristics of a **primary standard**.

19. NaOH is a chemical that can be obtained in pellet form. Explain why it is not used as a primary standard in volumetric analysis.

20. Complete the following table (for column five, assume that the acid is being added from a burette to a base in a conical flask).

Reactants (aqueous)	Salt Produced	Nature of solution at equivalence point	Suitable indicator	Expected colour change indicating end point
NaOH + HCl	NaCl	neutral	methyl orange or phenolphthalein	yellow to red pink to colourless
Na ₂ CO ₃ + HCl				
NH ₃ + HCl				
NaOH + CH ₃ COOH				

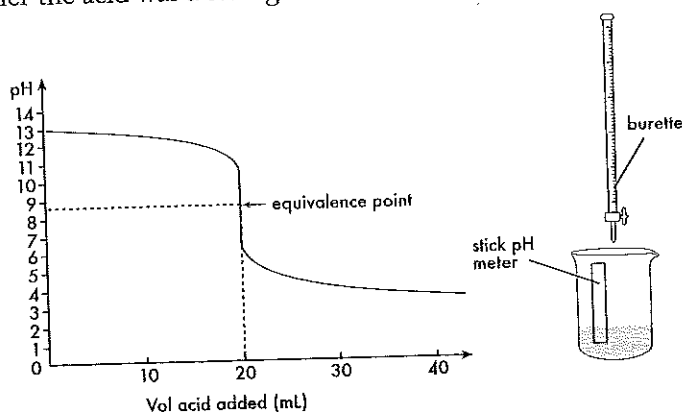
21. In determining the salt content of a sample of water a student added silver nitrate from a burette to 25.00 mL samples of the water in conical flasks. The student obtained the following results:

	Titration 1	Titration 2	Titration 3	Titration 4	Titration 5
Final Reading (mL)	21.00	41.50	26.50	46.90	23.40
Initial Reading (mL)	0.00	21.00	2.50	26.50	3.00
Volume Added (mL)	21.00	20.50	24.00	20.40	20.40

In calculating the average volume of AgNO₃ used in the titration, the student left the readings for titrations 1 and 3 out of the calculation. Explain why each was not included in the calculation of the average titre volume.

22. In the determination of the ammonia concentration of a window cleaning solution, a student used a previously standardised solution of hydrochloric acid. The student was unsure about what indicator to use and so chose to carry out two sets of titrations, one using methyl orange as the indicator and the other using phenolphthalein. The average volume of HCl solution used to neutralise the ammonia was 34.2 mL as indicated by the methyl orange. In general terms what would you expect the titre volume to be when phenolphthalein was used? Explain your answer.

23. Rather than use an indicator, a student used a pH meter to measure the variation of pH during a titration involving 0.0500 mol L⁻¹ NaOH and 0.0200 mol L⁻¹ of a certain acid. A graph of the titration results is given below. From the graph, state whether the acid was a strong acid or a weak acid. Explain your answer.



24. During a titration, a student noticed that some of the solution in the conical flask had splashed up and adhered to the inside region of the conical flask. Concerned that having some of the solution sitting above the remainder of the solution would ruin the results, the student used distilled water to wash the droplets back down into the main solution. Has this procedure altered the accuracy of the titration results?
25. In determining the ethanoic acid content of vinegar or the ammonia content of window cleaner, the household products are diluted by a factor of at least 10 before they are used in titrations. Explain the reason for the use of diluted solutions.

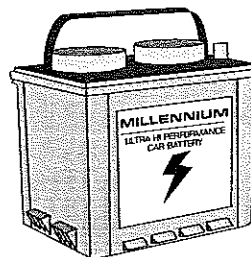
Calculations

26. A task carried out by high school laboratory assistants is the production of low concentration acid solutions for use in a variety of experiments. It is usual for these solutions to be produced from concentrated stock solutions. The three commonly prepared acids in school are made from $18.0 \text{ mol L}^{-1} \text{H}_2\text{SO}_4$, $14.0 \text{ mol L}^{-1} \text{HNO}_3$ or $12.0 \text{ mol L}^{-1} \text{HCl}$. Calculate the volume of stock acid that a lab assistant would need to obtain to make up each of the following dilute solutions:

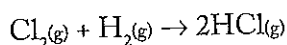
- (a) 250 mL of $2.00 \text{ mol L}^{-1} \text{H}_2\text{SO}_4$
- (b) 500 mL of $0.100 \text{ mol L}^{-1} \text{HNO}_3$
- (c) 0.750 L of $0.500 \text{ mol L}^{-1} \text{HCl}$

27. Upon reading the instructions for the initial setting up of a lead acid accumulator, a motorist found that they needed to obtain 2.00 L of 35.0% (by mass) H_2SO_4 . This acid was stated as having a density of 1.224 g mL^{-1} .

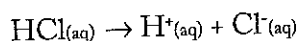
Calculate the number of moles of H_2SO_4 that must be added to each of the six cells in the accumulator, assuming that each cell receives an equal share of the 2.00 L of solution.



28. To effectively protect against harmful bacteria and viruses, the pH of a swimming pool must be kept between 7.2 and 7.8. (The lower the pH the more effective the control but below 7.2 the conditions become uncomfortable for the swimmer.) If a swimming pool held $3.50 \times 10^4 \text{ L}$ of water, calculate:
- (a) the minimum acceptable hydrogen ion concentration;
 - (b) the number of hydronium ions that must be present in the swimming pool when the pH is 7.8;
 - (c) the number of moles of HCl that would need to be added to the swimming pool to change the pH from 7.8 to 7.2;
 - (d) the volume of concentrated HCl (12 mol L^{-1}) that would need to be added to the pool to achieve this change in pH.
29. Hydrogen chloride gas can be produced by burning chlorine gas in hydrogen gas.



When this very soluble gas dissolves in water it completely ionises.

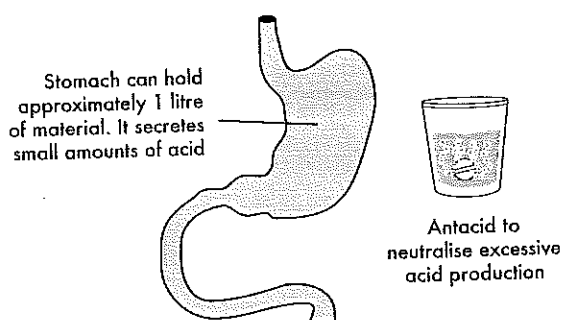


The entire contents of a 15.0 L cylinder of Cl_2 was burnt in excess H_2 and the resulting gas was dissolved in enough water to form 1.00 L of solution.

- If the chlorine gas cylinder was kept at a temperature of 25.0°C and the gas pressure was 1120 kPa, calculate the concentration of the HCl solution formed.
- The solution formed was found to have a density of 1150 g L^{-1} , calculate the percentage (by mass) of HCl in the solution.

30. A 375 mL bottle of antacid was analysed and found to contain 30.0 g of $\text{Mg}(\text{OH})_2$ and 30.0 g of $\text{Al}(\text{OH})_3$. The instructions on the bottle recommend a dose of 10.0 mL of the antacid.

Calculate the quantity of stomach acid that could be neutralised by taking the recommended dosage of antacid. Stomach acid has a pH of 0.900.



31. 25.3 g of sodium hydroxide was mixed with 9.60 g of magnesium chloride and then placed into enough water to produce 8.45 L of solution. Calculate:

- the $[\text{OH}^-]$ ion concentration of the resulting solution;
- the $[\text{H}^+]$ ion concentration of the resulting solution;
- the pH of the resulting solution.

32. The standardisation of an approximately 0.05 mol L^{-1} HCl solution with 20.0 mL aliquots of a 0.0997 mol L^{-1} Na_2CO_3 solution produced the following table of results:

	Titration Number				
	Rough	1	2	3	4
Final Reading (mL)	45.6	43.00	46.40	44.80	49.30
Initial Reading (mL)	2.0	1.00	5.00	3.50	7.90
Titration Volume (mL)					

- Write a balanced equation for the reaction.
- Calculate the number of moles of Na_2CO_3 used in each titration.
- Determine the average litre volume of HCl used.
- Determine the concentration of the HCl solution.

33. The sodium hydroxide concentration of an oven cleaning solution was determined by the steps outlined below.
- I 25.0 g of the oven cleaner was dissolved in 100 mL of distilled water.
 - II This solution was then added to a 250.0 mL volumetric flask and diluted by the addition of enough distilled water to correctly fill the flask.
 - III A 24.0 mL aliquot was removed from the volumetric flask and diluted by addition to enough water to fill another 250.0 mL volumetric flask.
 - IV 20.0 mL samples of this diluted solution were titrated against a standardised solution of HCl. An average of 29.8 mL of the 0.107 mol L^{-1} HCl was used.
- (a) Assuming that the only basic substance in the oven cleaner was sodium hydroxide, determine the sodium hydroxide concentration of the diluted solution produced in step III.
 - (b) Calculate the percentage, by mass, of NaOH in the original oven cleaning solution.
34. Car battery acid contains approximately $4.5 \text{ mol L}^{-1} \text{ H}_2\text{SO}_4$. A student carried out a series of titrations to determine the accurate concentration of H_2SO_4 in battery acid. The student followed the following steps.
- STEP I Take 20.0 mL of the battery acid and dilute to 1.00 L in a volumetric flask.
- STEP II Fill a burette with the diluted battery acid.
- STEP III Place 25.0 mL of standardised Na_2CO_3 ($0.0939 \text{ mol L}^{-1}$) into a conical flask and add several drops of methyl orange.
- STEP IV Conduct enough titrations to get results which are consistent to $\pm 0.1 \text{ mL}$.
- Calculate the concentration of the undiluted battery acid if the average titration volume obtained in step IV was 24.7 mL.
35. When the equivalence point in a titration between a standardised 0.100 mol L^{-1} HCl solution and an approximately 0.10 mol L^{-1} NaOH has been reached, 24.9 mL of the acid had been added from a burette to 20.0 mL of NaOH in a conical flask. What would be the change in pH caused by the addition of another 0.100 mL drop of acid into the solution in the conical flask?

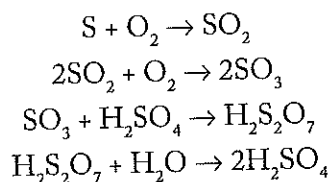
Equations for Acid Base Reactions

The writing of correctly balanced equations is often a major hurdle that stops many students beginning a calculation. See chapter 8 for further practice.

For the Experts

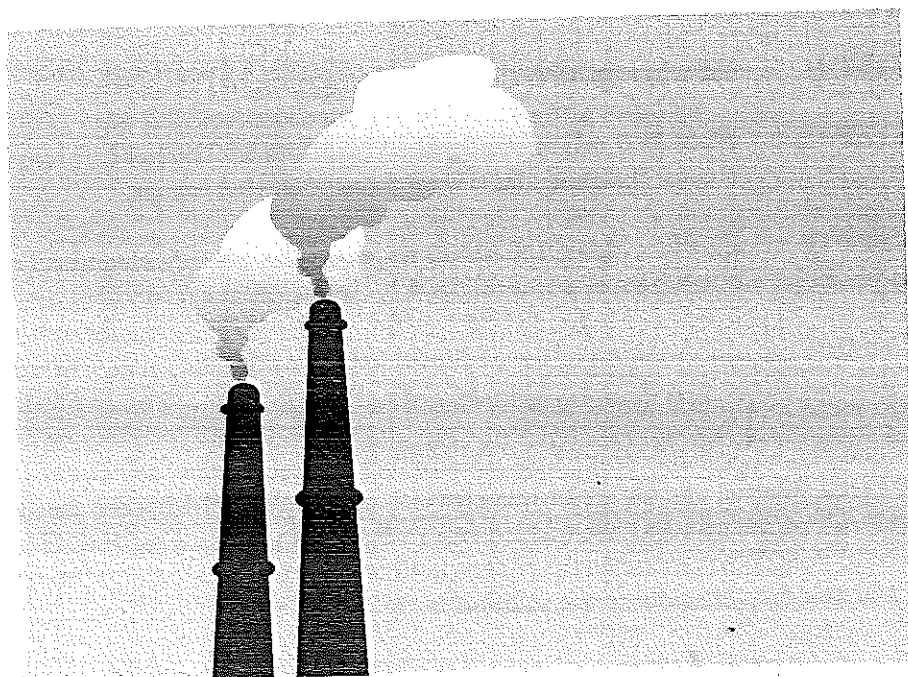
RESOURCE CHEMISTRY

36. As the world's supplies of fossil fuels dwindle and energy demands increase, there will be growing pressure to use more coal. An environmental issue that this raises is the production of SO_2 when coal is burnt. A challenge for chemical engineers is to find improved methods of removing sulfur from coal before it is burnt. A major use of this sulfur could be in the production of sulfuric acid:



Australia produces 420 million tonnes of coal annually that has an average sulfur content of 0.800% (w/w). If the recovery rate of sulfur from the coal is 85.0%, calculate:

- (a) The mass of sulfuric acid that could be produced from sulfur removed from the coal.
- (b) The mass of sulfur dioxide that would be added to the atmosphere when the sulfur reduced coal is burnt to produce energy.



$$\% \text{ mass} = \frac{6.73 \times 10^{-1} \text{ g}}{15.50} \times \frac{100}{1} = 4.34 \%$$

$$\begin{aligned} (c) \quad n(\text{CH}_3\text{COOH}) &= n(\text{CH}_3\text{COOH}) \\ &\text{in 250 mL dilute} \quad \text{in 15.50 mL commercial} \\ &= 1.120 \times 10^{-2} \\ c(\text{CH}_3\text{COOH}) &= \frac{1.120 \times 10^{-2}}{0.01550} \\ (i) &= 7.23 \times 10^{-1} \text{ mol L}^{-1} \\ (ii) &= 43.4 \text{ g L}^{-1} \end{aligned}$$

5.17

$$(a) \quad M(\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}) = 126 \text{ g mol}^{-1}$$

$$\begin{aligned} n(\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}) &= \frac{m}{M} = \frac{0.554}{126} \\ &= 4.40 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} c(\text{H}_2\text{C}_2\text{O}_4(\text{aq})) &= \frac{n}{V} = \frac{4.40 \times 10^{-3}}{0.500} \\ &= 8.79 \times 10^{-3} \text{ mol L}^{-1} \end{aligned}$$

$$\begin{aligned} (b)(i) \quad &\text{phenolphthalein} \\ (ii) \quad &\text{pink} \rightarrow \text{colourless} \\ (iii) \quad &2\text{NaOH}(\text{aq}) + \text{H}_2\text{C}_2\text{O}_4(\text{aq}) \rightarrow \\ &2\text{H}_2\text{O}(\text{l}) + \text{Na}_2\text{C}_2\text{O}_4(\text{aq}) \\ n(\text{oxalic acid}) &= cV \\ &= 8.79 \times 10^{-3} \times 0.01965 \\ &= 1.73 \times 10^{-4} \text{ mol L}^{-1} \\ n(\text{NaOH}) &= 2n(\text{H}_2\text{C}_2\text{O}_4) = 3.45 \times 10^{-4} \\ c(\text{NaOH}) &= \frac{n}{V} = \frac{3.45 \times 10^{-4}}{0.0200} \\ &= 1.73 \times 10^{-2} \text{ mol L}^{-1} \end{aligned}$$

$$\begin{aligned} (c)(i) \quad &\text{phenolphthalein} \\ (ii) \quad &\text{pink} \rightarrow \text{colourless} \\ (iii) \quad &n(\text{NaOH}) = c \times V \\ &= 1.73 \times 10^{-2} \times 0.0200 \\ &= 3.46 \times 10^{-4} \text{ mol} \\ c(\text{CH}_3\text{COOH}) &= n(\text{NaOH}) \\ &= 3.46 \times 10^{-4} \text{ mol} \\ c(\text{CH}_3\text{COOH}) &= \frac{n}{V} = \frac{3.46 \times 10^{-4}}{0.03250} \\ &= 1.06 \times 10^{-2} \text{ mol L}^{-1} \end{aligned}$$

5. Review Questions

1. Test the electrical conductivity of each solution. Comparatively, the strong electrolyte will be the best conductor, then the weak electrolyte and the non-electrolyte should not conduct.

2. STRONG	WEAK	NON
Na_2CO_3	NH_3	$\text{CH}_3\text{CH}_2\text{OH}$
HCl	CH_3COOH	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$

$$\begin{aligned} &\text{NaCl} \quad \text{SO}_2 \\ 3. (a) \quad &[\text{H}^+] = 1.00 \times 10^{-4} \text{ mol L}^{-1} \\ &[\text{OH}^-] = 1.00 \times 10^{-10} \text{ mol L}^{-1} \\ (b) \quad &[\text{H}^+] = 1.00 \times 10^{-10} \text{ mol L}^{-1} \\ &[\text{OH}^-] = 1.00 \times 10^{-4} \text{ mol L}^{-1} \\ (c) \quad &[\text{H}^+] = 3.50 \times 10^{-6} \text{ mol L}^{-1} \\ &[\text{OH}^-] = 2.86 \times 10^{-9} \text{ mol L}^{-1} \\ (d) \quad &[\text{H}^+] = 1.05 \times 10^{-13} \text{ mol L}^{-1} \\ &[\text{OH}^-] = 9.50 \times 10^{-2} \text{ mol L}^{-1} \end{aligned}$$

$$4. (a) 3 \quad (b) 5 \quad (c) 12.6 \quad (d) 3.52 \quad (e) 9.18$$

5.

Material	pH	$[\text{H}^+]$ (mol L ⁻¹)	$[\text{OH}^-]$ (mol L ⁻¹)
vinegar	3.00	1.00×10^{-3}	1.00×10^{-11}
toothpaste	6.80	1.58×10^{-7}	6.30×10^{-8}
oven cleaner	13.5	3.16×10^{-14}	0.316
window cleaner	9.75	1.79×10^{-10}	5.6×10^{-5}

$$\begin{aligned} 6. \quad c(\text{HCl}) &= 3.16 \times 10^{-5} \text{ mol L}^{-1} \\ V(\text{HCl}) &= 0.120 \text{ L} \\ n(\text{HCl}) &= 3.795 \times 10^{-6} \\ c(\text{NaOH}) &= 1.58 \times 10^{-4} \text{ mol L}^{-1} \\ V(\text{NaOH}) &= 0.075 \text{ L} \\ n(\text{NaOH}) &= 1.189 \times 10^{-5} \end{aligned}$$

$$\begin{aligned} &\text{HCl completely consumed;} \\ &n(\text{NaOH}) \text{ remaining} \\ &= 1.18 \times 10^{-5} - 3.792 \times 10^{-6} \\ &= 8.092 \times 10^{-6} \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{ie. } [\text{OH}^-] &= \frac{8.092 \times 10^{-6}}{(0.120 + 0.075)} \\ &= 4.15 \times 10^{-5} \text{ mol L}^{-1} \end{aligned}$$

$$\begin{aligned} [\text{H}^+] &= \frac{1.00 \times 10^{-14}}{4.15 \times 10^{-5}} \\ &= 2.41 \times 10^{-10} \text{ mol L}^{-1} \\ \text{pH} &= 9.62 \end{aligned}$$

7. CH_3COOH is a weak acid and will only partially ionise in solution (producing H_3O^+) and so will produce a solution with a higher pH than HNO_3 or HCl which are strong acids.

8. (a) It will be considered neutral if $[\text{H}^+] = [\text{OH}^-]$. In this example $[\text{H}^+][\text{OH}^-] \neq 1 \times 10^{-14}$ – a possible explanation is that the temperature of the solution was not at 25°C .
(b) The temperature would be less than 25°C .
(c) If neutral $K_w = [\text{H}^+]^2 = 2.86 \times 10^{-15}$.
(d) For solution temperatures greater than 25°C .

$$9. (a) \text{ Original pH} = 7.8, V = 45,000 \text{ L}$$

$$[H^+] = 1.58 \times 10^{-8} \text{ mol L}^{-1}$$

$$\text{original } n(H^+) = c \cdot V = 1.58 \times 10^{-8} \times 45,000 = 7.13 \times 10^{-4}$$

$$m(HCl) \text{ added} = 0.900 \times 320 = 288 \text{ g}$$

$$n(HCl) \text{ added} = m/M = 288/(1.008 + 35.45) = 7.90$$

$$\text{new conc} = \frac{n \text{ old} + n \text{ added}}{V \text{ old} + V \text{ added}} = \frac{7.13 \times 10^{-4} + 7.90}{45,000}$$

$$c(HCl) = 1.76 \times 10^{-4}$$

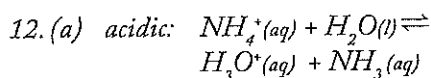
$$pH = 3.76$$

(b) The pool contained a large amount of buffer solution compared to amount of HCl added.

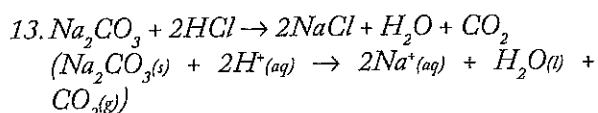
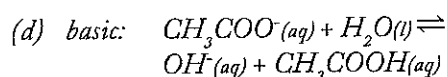
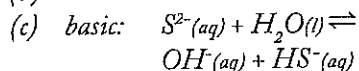
10. (a) NH_4^+ and NH_3 OH^- and H_2O
 (b) HPO_4^{2-} and $H_2PO_4^-$ H_2CO_3 and HCO_3^-
 (c) HCO_3^- and H_2CO_3 H_3O^+ and H_2O
 (d) CH_3COOH and CH_3COO^- NH_3 and NH_4^+

11.

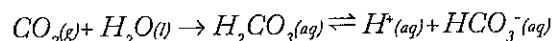
Reactants	Salt Formed	Nature of the salt
$HCl + NH_3$	NH_4Cl	acidic
$H_2SO_4 + KOH$	K_2SO_4	slightly basic
$CH_3COOH + NaOH$	$NaCH_3COO$	basic
$HNO_3 + Mg(OH)_2$	$Mg(NO_3)_2$	neutral
$H_3PO_4 + NaOH$	Na_3PO_4	basic



(b) neutral

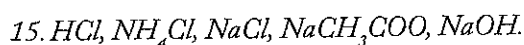


The resulting solution is acidic because CO_2 dissolves in water to form solution of the weak acid H_2CO_3



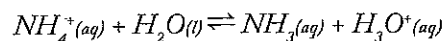
14.

- (a) Add $H_2SO_4(aq)$ to both: the solution which produced a white precipitate would be the $Ba(OH)_2$.
 (b) Dissolve both in water and test pH (with an indicator or a meter), the neutral solution would be the $NaCl$; the slightly basic solution would be the sodium ethanoate.
 (c) Add $Ba(NO_3)_2(aq)$ to both: a white precipitate will form in the sulfuric acid.

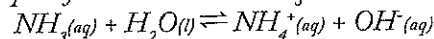


16.

(a) pH of solution A < 7; hydrolysis of $NH_4^+(aq)$



pH of solution B > 7; NH_3 is a weak base



- (b) Not expected to separate $[NH_3]$ and $[NH_4^+]$; but their concentrations will be greatest and concentration of H_3O^+ and OH^- will be least.
 (c) $[NH_3] > [OH^-]$, therefore H^+ ions more likely to collide and react with NH_3 .
 (d) (i) H^+ ions would react with OH^- ions, forward reaction is favoured to partially counteract the reduced $[OH^-]$.
 (ii) The $[OH^-]$ will increase and the reverse reaction will be favoured to partially counteract this change.
 In both cases, the pH will remain very close to 9.
 (e) The buffer capacity could be exceeded by adding sufficiently large amounts of H^+ or OH^- ions that the buffer could not counteract.

17.

- (a) Primary Standard: substance that can be used to produce a pure solution of accurately known concentration.
 (b) Standardisation: when a solution of known concentration is used to accurately determine the concentration of another solution.
 (c) End Point: point at which the titration is stopped because the desired colour change is observed.
 (d) Equivalence Point: point at which stoichiometrically equivalent amounts of reactants have been added to the reaction vessel.
 (e) Indicator: substance which has a definite colour change when subject to different pH's.
 (f) Rough Titration: first titration that is done quickly to get an approximate quantity of reactant needed to reach the end point.

18. (a) Should be obtained in a very pure form.
 (b) Have a known formula.
 (c) Have a high molar mass.
 (d) Should not react with air or absorb moisture from the air.

19. It is difficult to obtain pure as it is deliquescent - ie. it absorbs moisture from the air and can form a solution in this moisture.

20.

Reactant	Salt Produced	Nature of soln at equiv. point	Suitable indicator	Expected colour change
$Na_2CO_3 + HCl$	$NaCl$	acidic (due to CO_2)	methyl orange	yellow to red
$NH_3 + HCl$	NH_4Cl	acidic	methyl orange	yellow to red
$NaOH + CH_3COOH$	$NaCH_3COO$	basic	phenolphthalein	pink to colourless

21. Titration 1: should be a rough titration only.
 Titration 3: result is outside range of precision of ± 0.10 mL, titrations 2, 4 and 5.

22. $\text{HCl} + \text{NH}_3$: end point will be in acidic region. Methyl orange should be used as an indicator. If phenolphthalein is used then the volume of HCl added will be less as the phenolphthalein will change at a pH of approximately 8 rather than 4.

23. Graph indicates an equivalence point in the basic region, indicative of a titration involving a strong base and a weak acid.

24. No as it has not changed the number of moles of solution in the conical flask.

25. Indicator changes colour more slowly and larger volumes are used which give smaller percentage errors. Burette would not need to be continually refilled.

26.

$$(a) n(\text{conc. H}_2\text{SO}_4) = n(\text{dilute H}_2\text{SO}_4)$$

$$c_1 V_1 = c_2 V_2$$

$$V_1 = \frac{c_2 V_2}{c_1} = \frac{0.250 \times 2.00}{18.0}$$

$$= 0.0278 \text{ L}$$

$$V_1(\text{conc. H}_2\text{SO}_4) = 27.8 \text{ mL}$$

$$(b) V(\text{conc. HNO}_3)$$

$$= \frac{c_2 V_2}{c_1} = \frac{0.500 \times 0.100}{14.0} = 0.00357 \text{ L}$$

$$V(\text{conc. HNO}_3) = 3.57 \text{ mL}$$

$$(c) V(\text{conc. HCl})$$

$$= \frac{c_2 V_2}{c_1} = \frac{0.750 \times 0.500}{12.0} = 0.0312 \text{ L}$$

$$V(\text{conc. HCl}) = 31.2 \text{ mL}$$

27.

$$(a) m(\text{battery acid}) = D \times V$$

$$= 1.224 \times 2.00 \times 10^3 = 2440 \text{ g}$$

$$m(\text{H}_2\text{SO}_4) = 35.0\% \text{ of } m(\text{battery acid})$$

$$= 2440 \times \frac{35.0}{100} = 854 \text{ g}$$

$$n(\text{H}_2\text{SO}_4) = \frac{m}{M} = \frac{854}{98.086} = 8.71 \text{ mol}$$

$$n(\text{H}_2\text{SO}_4) \text{ per cell} = 8.71 \div 6$$

$$= 1.45 \text{ mol}$$

28.

$$(a) \min [\text{H}^+] = \text{inv log } (-7.8)$$

$$= 1.58 \times 10^{-8} \text{ mol L}^{-1}$$

$$(b) \text{ no. H}_3\text{O}^+ \text{ ions} = c \times V \times 6.022 \times 10^{23}$$

$$= 1.58 \times 10^{-8} \times 3.50 \times 10^4 \times 6.022 \times 10^{23}$$

$$= 3.34 \times 10^{20} \text{ ions}$$

$$(c) \text{ If pH} = 7.2, [\text{H}^+] = \text{inv log } (-7.2)$$

$$= 6.31 \times 10^{-8} \text{ mol L}^{-1}$$

$$n(\text{acid}) \text{ in pool if pH} = 7.8 = cV$$

$$= 1.58 \times 10^{-8} \times 3.50 \times 10^4 = 5.53 \times 10^{-4} \text{ mol}$$

$$n(\text{acid}) \text{ in pool if pH} = 7.2 = cV$$

$$= 6.31 \times 10^{-8} \times 3.50 \times 10^4 = 2.21 \times 10^{-3} \text{ mol}$$

$$n(\text{HCl}) \text{ to be added}$$

$$= 2.21 \times 10^{-3} - 5.53 \times 10^{-4}$$

$$= 1.657 \times 10^{-3} \text{ mol}$$

$$(d) V(\text{conc. HCl}) = \frac{n}{c} = \frac{1.657 \times 10^{-3}}{12}$$

$$= 1.38 \times 10^{-4} \text{ L (0.138 mL)}$$

29.

$$(a) n(\text{Cl}_2) = \frac{PV}{RT} = \frac{1120 \times 15.0}{8.315 \times 298.1} = 6.78 \text{ mol}$$

$$(\text{dissolved in 1.00 L of water})$$

$$n(\text{HCl}) = 2n(\text{Cl}_2) = 13.6 \text{ mol}$$

$$c(\text{HCl}) = 13.6 \text{ mol L}^{-1}$$

$$(b) m(\text{HCl}) = nM = 13.6 \times 36.458$$

$$= 494 \text{ g}$$

$$m(\text{soln}) = 1150 \text{ g}$$

$$\% \text{ HCl} = \frac{494}{1150} \times \frac{100}{1} = 43.0\%$$

$$30. n(\text{Mg(OH)}_2) = \frac{m}{M} = \frac{30.0}{58.316} = 0.514 \text{ mol}$$

$$n(\text{OH}^-) = 2 \times 0.514 = 1.029$$

$$n(\text{Al(OH)}_3) = \frac{m}{M} = \frac{30.0}{78.004} = 0.385 \text{ mol}$$

$$n(\text{OH}^-) = 3 \times 0.385 = 1.154 \text{ mol}$$

$$n(\text{OH}^-) \text{ in 375 mL} = 1.029 + 1.154 = 2.18 \text{ mol}$$

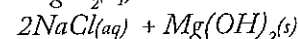
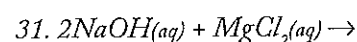
$$n(\text{OH}^-) \text{ in 10.0 mL} = 2.18 \times \frac{10}{375} = 0.0582 \text{ mol}$$

$$\therefore n(\text{HCl}) \text{ neutralised} = n(\text{OH}^-) \text{ in 10.0 mL}$$

$$= 0.0582$$

$$V(\text{HCl}) \text{ neutralised}$$

$$= \frac{n}{c} = \frac{0.0582}{\text{inv log } (-0.9)} = 0.462 \text{ L}$$



$$n(\text{NaOH}) = \frac{m}{M} = \frac{25.3}{39.998} = 0.633 \text{ mol}$$

$$n(\text{MgCl}_2) = \frac{m}{M} = \frac{9.60}{95.2} = 0.101 \text{ mol}$$

$$(\text{Limiting Reagent})$$

$$n(\text{OH}^-) \text{ consumed} = 2n(\text{MgCl}_2) = 0.202 \text{ mol}$$

(a) $n(\text{OH}^-)$ remaining in solution

$$= 0.633 - 0.202 = 0.431 \text{ mol}$$

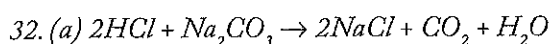
$c(\text{OH}^-)$ remaining in solution

$$= \frac{n}{V} = \frac{0.431}{8.45} = 0.0510 \text{ mol L}^{-1}$$

$$(b) c(\text{H}^+) = \frac{1.00 \times 10^{-14}}{c(\text{OH}^-)} = \frac{1.00 \times 10^{-14}}{0.0510}$$

$$= 1.96 \times 10^{-13} \text{ mol L}^{-1}$$

$$(c) \text{pH} = -\log [\text{H}^+] = 12.7$$



$$(b) n(\text{Na}_2\text{CO}_3) = cV = 0.997 \times 0.0200$$

$$= 1.994 \times 10^{-3} \text{ mol}$$

(c) average $V(\text{HCl}) =$

$$(\text{titration 2} + \text{titration 3} + \text{titration 4}) \div 3$$

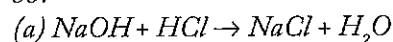
$$= 41.4 \text{ mL}$$

$$32. (d) n(\text{HCl}) = 2n(\text{Na}_2\text{CO}_3) = 3.988 \times 10^{-3} \text{ mol}$$

$$c(\text{HCl}) = \frac{n}{V} = \frac{3.99 \times 10^{-3}}{0.0414}$$

$$= 0.0963 \text{ mol L}^{-1}$$

33.



$$n(\text{HCl}) = cV = 0.107 \times 0.0298$$

$$n(\text{HCl}) = 3.189 \times 10^{-3} \text{ mol}$$

$$n(\text{NaOH}) = n(\text{HCl}) = 3.189 \times 10^{-3} \text{ mol}$$

$$c(\text{dilute NaOH}) = \frac{n}{V} = \frac{3.189 \times 10^{-3}}{0.0200}$$

$$= 0.159 \text{ mol L}^{-1}$$

(b) (Oven cleaner was diluted twice: in steps II and III)

$n(\text{NaOH})$ in 250 mL of dilute solution from step III

$$= cV = 0.159 \times 0.250$$

$n(\text{NaOH})$ in 250 mL from step III

$$= 0.0399 \text{ mol}$$

$\therefore n(\text{NaOH})$ in 24.0 mL from step II

$$= 0.0399 \text{ mol}$$

$$c(\text{NaOH}) \text{ from step II} = \frac{0.0399}{0.0240}$$

$$= 1.66 \text{ mol L}^{-1}$$

$\therefore n(\text{NaOH})$ in all of solution in step II

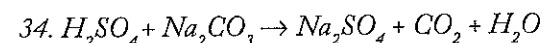
$$= cV = 1.66 \times 0.250 = 0.415 \text{ mol}$$

$m(\text{NaOH})$ in all of solution in step II

$$= nM = 0.415 \times 39.998 = 16.6 \text{ g}$$

% NaOH in drain cleaner

$$= \frac{16.6}{25.0} \times \frac{100}{1} = 66.4\% (\text{w/w})$$



$$n(\text{Na}_2\text{CO}_3) = cV = 0.0939 \times 0.025 = 2.35 \times 10^{-3}$$

$$\therefore n(\text{dilute H}_2\text{SO}_4) = n(\text{Na}_2\text{CO}_3) = 2.35 \times 10^{-3}$$

$$c(\text{dilute H}_2\text{SO}_4) = \frac{n}{V} = \frac{2.35 \times 10^{-3}}{0.0247}$$

$$= 0.0950 \text{ mol L}^{-1}$$

$$n(\text{dilute H}_2\text{SO}_4) \text{ in } 1.00 \text{ L} = cV$$

$$= 0.0950 \times 1.00 = 0.0950 \text{ mol}$$

$$\therefore n(\text{conc. H}_2\text{SO}_4) \text{ in } 20.0 \text{ mL}$$

$$= n(\text{dilute H}_2\text{SO}_4) \text{ in } 1.00 \text{ L}$$

$$= 0.0950 \text{ mol}$$

$$c(\text{battery acid}) = \frac{n}{V} = \frac{0.0950}{0.0200}$$

$$= 4.75 \text{ mol L}^{-1}$$

35. At equivalence point the solution is neutral

$$\text{ie. } [\text{H}^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$$

$$V(\text{solution}) = 0.0249 + 0.0200 = 0.0449 \text{ L}$$

$$\therefore n(\text{H}^+) \text{ at equivalence point} = cV$$

$$= 1.00 \times 10^{-7} \times 0.0449 = 4.49 \times 10^{-9} \text{ mol}$$

$$n(\text{H}^+) \text{ in 1 drop} = cV = 0.100 \times 0.0001$$

$$= 1.00 \times 10^{-5} \text{ mol}$$

$\therefore n(\text{H}^+) \text{ in solution after 1 drop added}$

$$= 4.49 \times 10^{-9} + 1.00 \times 10^{-5} = 1.00 \times 10^{-5} \text{ mol}$$

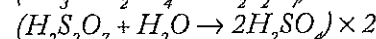
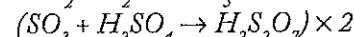
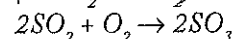
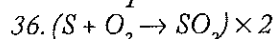
$$c(\text{H}^+) = \frac{n}{V} = \frac{1.00 \times 10^{-5}}{0.0449}$$

$$= 2.23 \times 10^{-4} \text{ mol L}^{-1}$$

$$\text{new pH} = -\log (2.23 \times 10^{-4}) = 3.65$$

$$\therefore \text{change in pH} = 7.00 - 3.65 = 3.35$$

For the Experts



$$(a) n(\text{H}_2\text{SO}_4) = 2n(\text{s})$$

$$m(\text{S}) \text{ in coal} = \frac{0.800}{100} \times 420 \times 10^{12}$$

$$= 3.36 \times 10^{12} \text{ g}$$

$$m(\text{S}) \text{ recovered} = 3.36 \times 10^{12} \times \frac{85}{100}$$

$$= 2.86 \times 10^{12} \text{ g}$$

$$n(\text{S}) \text{ recovered} = \frac{m}{M} = \frac{2.86 \times 10^{12}}{32.06}$$

$$= 8.91 \times 10^{10} \text{ mol}$$

$$n(\text{H}_2\text{SO}_4) = 2n(\text{S}) = 1.78 \times 10^{11} \text{ mol}$$

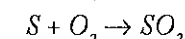
$$m(\text{H}_2\text{SO}_4) = nM = 1.78 \times 10^{11} \times 98.076$$

$$m(\text{H}_2\text{SO}_4) = 1.75 \times 10^{13} \text{ g}$$

$$(17.5 \text{ million tonnes})$$

$$(b) m(\text{S}) \text{ not recovered} = 3.36 \times 10^{12} \times \frac{15}{100}$$

$$= 5.04 \times 10^{11} \text{ g}$$



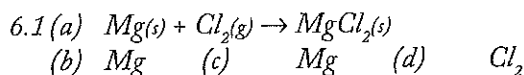
$$n(\text{SO}_2) = n(\text{S})$$

$$n(\text{S}) = \frac{m}{M} = \frac{5.04 \times 10^{11}}{32.06}$$

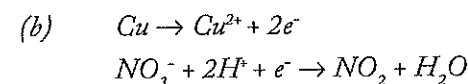
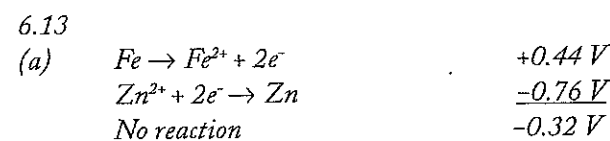
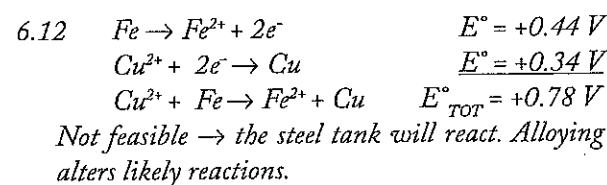
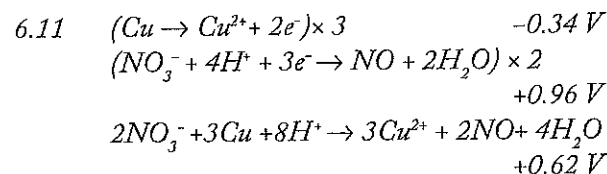
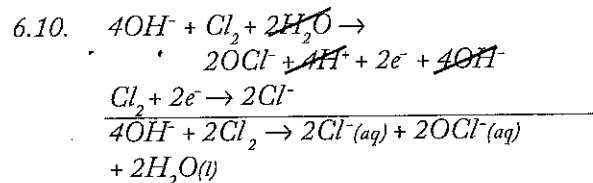
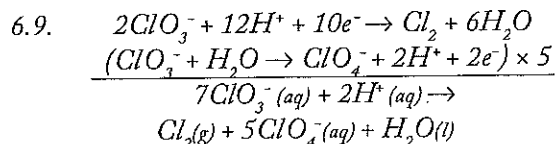
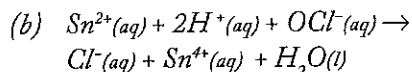
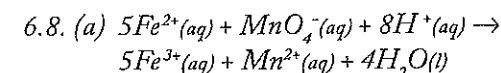
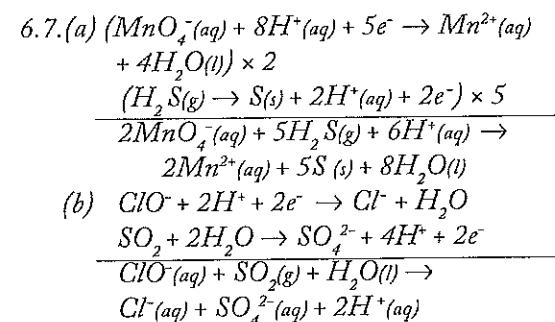
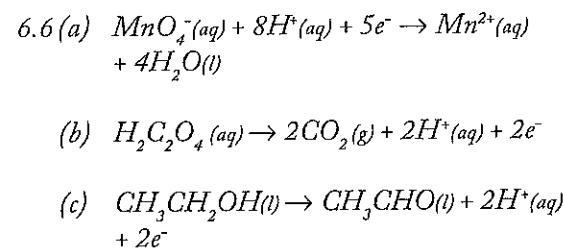
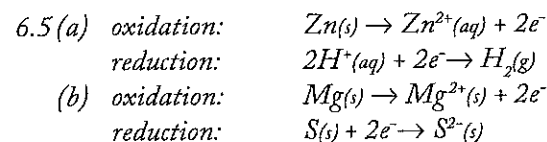
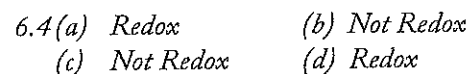
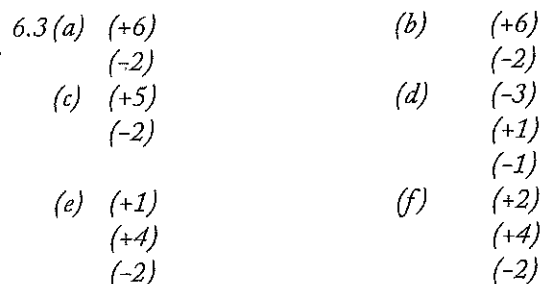
$$\begin{aligned}
 &= 1.57 \times 10^{10} \text{ mol} \\
 n(\text{SO}_2) &= 1.57 \times 10^{10} \text{ mol} \\
 m(\text{SO}_2) &= nM = 1.57 \times 10^{10} \times 64.06 \\
 m(\text{SO}_2) &= 1.01 \times 10^{12} \text{ g} \\
 &\quad (1.01 \text{ million tonnes})
 \end{aligned}$$

CHAPTER 6 - Oxidation & Reduction

6. Chapter Questions



6.2 True: oxidant removes electrons from reductant.
 eg. $2\text{Fe} + \text{O}_2 \rightarrow 2\text{FeO}$
 O_2 removes electrons from Fe to form O^{2-} and Fe^{2+} .



Reaction will occur since HNO_3 is concentrated.

